

Donkey Pluralities: Plural Information States vs. Non-atomic Individuals

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Plural Info State I	...	U	U'	...
i_1	...	x_1 (=jane $\oplus$ john)	y_1 (=mary)	...
i_2	...	x_2 (=bill $\oplus$ gaby)	y_2 (=sue $\oplus$ joe)	...
i_3	...	x_3 (=megan)	y_3 (=matt $\oplus$ rob)	...
...	...	...	...	...
Individuals, possibly non-atomic (plural reference): $\{x_1, x_2, x_3, \dots\}, \{y_1, y_2, y_3, \dots\}$ etc.		Quantificational dependencies between individuals (plural discourse reference): $\{<x_1, y_1>, <x_2, y_2>, <x_3, y_3>, \dots\}$ etc.		

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1. Introduction

The main goal of this presentation is to argue that two distinct and independent notions of plurality are involved in natural language anaphora and quantification:

- **plural reference**
- **plural discourse reference**

1. Introduction

Plural reference: the usual **non-atomic individuals**,

e.g. the non-atomic individual $megan \oplus gaby$ that is the sum of the two atomic individuals *megan* and *gaby* and that the sentence in (1) below is about.

1. Megan and Gaby are deskmates.

[*megan* and *gaby* are **atomic** individuals because they have no proper sub-parts that are individuals; $megan \oplus gaby$ is a **non-atomic** individual because it has such proper sub-parts.]

1. Introduction

Plural discourse reference: reference to a quantificational dependency between sets of atomic / non-atomic individuals that is established and subsequently elaborated upon in discourse,

e.g. the dependency between **gifts** and **girls** introduced in the first conjunct and elaborated upon in the second conjunct of sentence (2) below.

2. John bought **a^u gift** for **every^{u'} girl** in his class and asked **their_{u'}** deskmates to wrap **them_u**.

1. Introduction

2. John bought a^u gift for $\text{every}^{u'}$ girl in his class ...
... and asked their_u deskmates to wrap them_u .

[Indexation convention: superscripts – new referents are introduced;
subscripts – previously introduced referents are retrieved.

A $\langle \text{superscript}, \text{subscript} \rangle$ pair is basically a co-reference arrow.]



The first conjunct in (2) introduces a quantificational dependency between the set of **girls** in John's class and the set of **gifts** bought by John:

each **girl** is correlated with the **gift(s)** that John bought for her.

1. Introduction

2. John bought a^u gift for $every^{u'}$ girl in his class ...
... and asked $their_u$ deskmates to wrap $them_u$.

The second conjunct in (2) elaborates on the dependency between the previously mentioned girls and gifts:

for each girl, John asked her deskmate to wrap the corresponding gift(s).

1. Introduction

But:

morphologically plural anaphora of the kind instantiated in (2) does not provide a clear-cut argument for **plural discourse reference** over and above **plural reference**

(although much of the previous literature relies on such examples to argue for and talk about plural discourse reference):

both notions / either notion could be involved in the interpretation of (2).

1. Introduction

2. John bought a^u gift for $every^{u'}$ girl in his class and asked $their_{u'}$ deskmates to wrap $them_u$.

For example, it might be that:

- semantically, (2) involves only plural reference: two non-atomic individuals consisting of all the u -gifts and all the u' -girls
- the dependency between the atoms is only pragmatically supplied

which seems fairly plausible in view of examples like (3):

3. John bought a^u gift for $every^{u'}$ girl in his class and asked $them_{u'}$ / $the_{u'}$ girls to wrap $them_u$ / the_u gifts.

where the 'buying' correlation / dependency between the gift-atoms and the girl-atoms might very well be different from the 'wrapping' correlation / dependency.

1. Introduction

The plan.

- I will therefore use sentences with **multiple** instances of morphologically **singular** donkey anaphora to provide independent semantic motivation for plural discourse reference over and above plural reference
(unlike Rooth 1987, van den Berg 1996 and Krifka 1996 among many others)
- and I will use sentences with **plural** donkey anaphora to argue that we still need plural reference over and above plural discourse reference.

So: what is donkey anaphora?

2. Donkey Anaphora

... so-called because of the paradigm example of such anaphora (based on an example in Geach 1962):

4. Every^u farmer who owns **a^{u'} donkey** beats **it_{u'}**.

Here are two novel examples of **multiple singular** donkey anaphora (which we will examine in detail later on):

5. Every^u person who buys **a^{u'} book** on amazon.com and has **a^{u''} credit card** uses **it_{u''}** to pay for **it_{u'}**.

6. Every^u boy who bought **a^{u'} Christmas gift** for **a^{u''} girl in his class** asked **her_{u''}** deskmate to wrap **it_{u'}**.

2. Donkey Anaphora

Why is donkey anaphora interesting?

It is interesting from a **syntax/semantics interface** perspective:

- **semantically**: quantifier binding
- **syntactically**: no c-command

2. Donkey Anaphora

4. Every^{*u*} farmer who owns *a^{*u'*} donkey* beats *it_{*u'*}*.

The pronoun *it_{*u'*}* in (4) is not used referentially...

... at least not referentially in the sense of cross-sentential / cross-clausal anaphora of the kind instantiated in (7).

7. John bought *a^{*u'*} donkey*. *It_{*u'*}* hadn't been vaccinated.

2. Donkey Anaphora

4. Every^u farmer who owns a^{u'} donkey beats it_{u'}.
7. John bought a^{u'} donkey. It_{u'} hadn't been vaccinated.

In (7), it might be that simply saying the indef a^{u'} donkey evokes / brings to salience (pragmatically) ...

... some particular donkey  , namely the one that John bought on a particular occasion ...

... and then the pronoun it_{u'} is interpreted *referentially*:

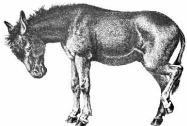
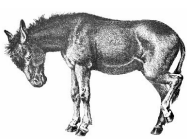
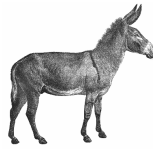
it_{u'} refers to this  (pragmatically) salient donkey.

2. Donkey Anaphora

4. Every^u farmer who owns a^{u'} donkey beats it_{u'}.
7. John bought a^{u'} donkey. It_{u'} hadn't been vaccinated.

But there is no salient donkey in (4), which is intuitively interpreted *quantificationally*, i.e. as:

every donkey owning farmer beats *every donkey* s/he owns.

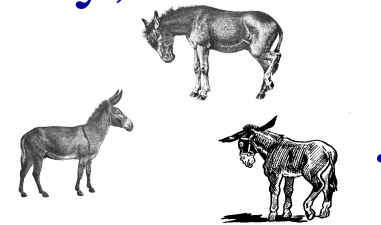
e.g., if John owns  ,  and  ,
then John beats  ,  and  .

2. Donkey Anaphora

4. Every^u farmer who owns **a^{u'} donkey** beats **it_{u'}**.

In particular, note that (4) does not intuitively apply only to farmers that own a single / unique donkey,

since farmer John has three donkeys:



Also, note that donkey anaphora receives a quantificational interpretation in episodic / non-generic sentences too:

8. Every^u farmer who bought **a^{u'} donkey** vaccinated **it_{u'}**.

2. Donkey Anaphora

Donkey anaphora contrasts with ordinary quantificational expressions, which cannot bind without c-command:

9. #Every^u boy who read *every^{u'} Harry Potter book* recommended *it_{u'}* to his_u friends.

10. Every^u boy who read *a^{u'} Harry Potter book* recommended *it_{u'}* to his_u friends.

2. Donkey Anaphora

The question that emerges is:

what is the notion of **reference** involved in natural language interpretation?

This notion of reference has to cover donkey anaphora (i.e. donkey indefinites and donkey pronouns)

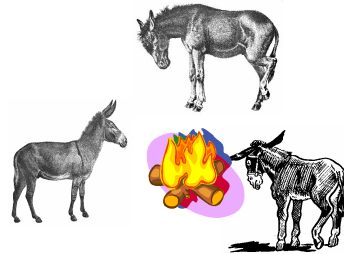
over and above run-of-the-mill referential expressions like *Megan* and ordinary quantificational expressions like *every book*.

2. Donkey Anaphora

Moreover, this notion of reference should also explain why **singular** donkey anaphora is incompatible with **collective** predicates:

11. #Every^u farmer who owns a^{u'} donkey gathers it_{u'} around the fire at night.
(based on an example in Kanazawa 2001)

despite the fact that farmer John has three donkeys and could very well gather them around the fire:



2. Donkey Anaphora

11. #Every^u farmer who owns a^{u'} donkey gathers it_{u'} around the fire at night.



... gathers it around the fire ...

2. Donkey Anaphora

11. #Every^u farmer who owns a^{u'} donkey gathers it_{u'} around the fire at night.



... gathers it around the fire ... like this?!

3. Dynamic Semantics

The kind of answer I will pursue for the question

What is the notion of **reference** involved in natural language interpretation?

is

Reference in natural language is inextricably **discourse reference**.

(Karttunen 1976, Kamp 1981, 2001 and Heim 1982 among many others)

3. Dynamic Semantics

"Consider a device designed to read a text in some natural language, interpret it and store the content in some manner, say, for the purpose of being able to answer questions about it.

"To accomplish this task, the machine will have to fulfill at least the following basic requirement.

"It has to be able to build a file that consists of records of all the individuals, that is, events, objects, etc., mentioned in the text and, for each individual, record whatever is said about it."

(Karttunen 1976)

3. Dynamic Semantics

"Consider a device designed to read a **TEXT** in some natural language, interpret it and store the content in some manner, say, for the purpose of being able to answer questions about it.

"To accomplish this task, the machine will have to fulfill at least the following basic requirement.

"It has to be able to build a **FILE** that consists of **RECORDS OF ALL THE INDIVIDUALS**, that is, events, objects, etc., mentioned in the text and, for each individual, record whatever is said about it."

(Karttunen 1976)

3. Dynamic Semantics

"Consider a device designed to read a **TEXT** ..."

... i.e. a **discourse**, not a sentence in isolation.

"As a rule discourse meanings are more than plain conjunctions of sentence meanings. And this 'more' is often the effect of interpretation principles that are an integral part of linguistic knowledge, and thus legitimate objects of linguistic study."

(Kamp 2001)

3. Dynamic Semantics

"Consider a device designed to read a **TEXT** ..."

The **information** that a sentence conveys is

NOT JUST: "information to the effect that the sentence is true, that the world is such as to meet the truth-conditions of the sentence." (Hintikka 1969)

BUT ALSO: information about how to **change** the **information** / **context** relative to which subsequent expressions are interpreted.

3. Dynamic Semantics

"Consider a device designed to read a **TEXT** ..."

The problem of discourse interpretation shifts our perspective on the **meaning** of natural language expressions

FROM: a static **truth-conditional** theory of meaning

TO: a dynamic theory of meaning as **information update**.

3. Dynamic Semantics

"[...] the machine [...] has to be able to build a **FILE** that consists of **RECORDS OF ALL THE INDIVIDUALS ...**"

We need a notion of **discourse reference** that mediates between linguistic expressions and their reference (in the classical, truth-conditional sense).

The text interpreter incrementally builds a **file**, which encodes the current **information state** of the interpreter, i.e. the **context of interpretation** (incrementally updated in discourse).

3. Dynamic Semantics

"[...] the machine [...] has to be able to build a **FILE** that consists of **RECORDS OF ALL THE INDIVIDUALS** ..."

The **information state** should be able to:

- encode when a novel individual is mentioned in discourse
- store it along with its characterization for future reference

Thus, minimally, information states are **sequences of individuals** (i.e. variable assignments) that satisfy certain characterizations (i.e. predicates).

3. Dynamic Semantics

"[...] the machine [...] has to be able to build a **FILE** that consists of **RECORDS OF ALL THE INDIVIDUALS ...**"

And the **information state**, i.e. the **sequence of individuals**, is incrementally updated in discourse in two ways:

- we expand it by adding new individuals
- we check that the individuals stored in the sequence satisfy whatever new properties we predicate of them

3. Dynamic Semantics

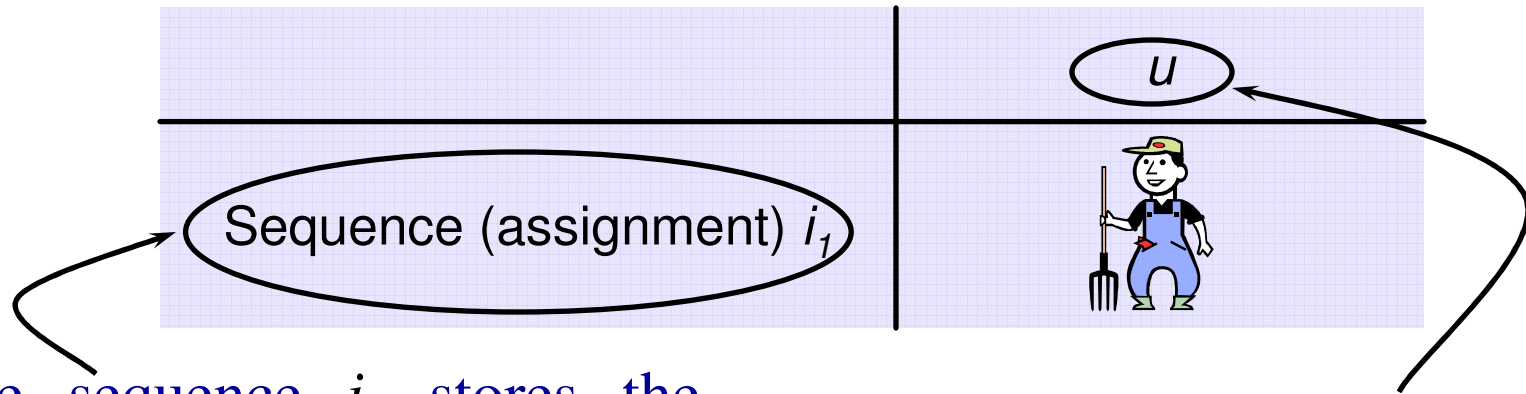
4. Every^{*u*} farmer who owns a^{*u'*} donkey beats it_{*u'*}.

Left-to-right incremental update:

- every^{*u*} farmer instructs us to introduce a farmer relative to the discourse referent (dref) *u*

	<i>u</i>
Sequence (assignment) <i>i</i> ₁	

3. Dynamic Semantics



- the sequence i_1 stores the association between the tagged pigeonhole u and the (representation of) the individual stored relative to u ;
- that is, i_1 is an entire set of tagged pigeonholes together with their associated contents

- the dref u is the index / 'handle' relative to which we introduce various (representations of) individuals and by means of which we can subsequently access them;
- think of it as a pigeonhole tagged with u

3. Dynamic Semantics

4. Every^{*u*} farmer who owns *a^{u'} donkey* beats *it_{u'}*.

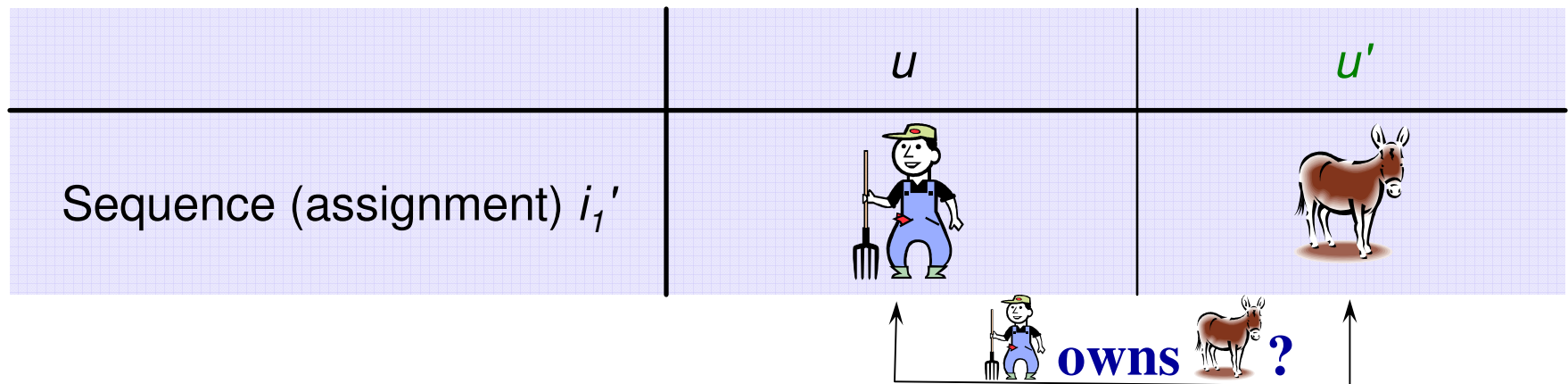
- *a^{u'} donkey* instructs us to introduce a donkey relative to the dref *u'*

	<i>u</i>	<i>u'</i>
Sequence (assignment) <i>i₁'</i>		

3. Dynamic Semantics

4. Every^{*u*} farmer who owns a^{*u'*} donkey beats it_{*u'*}.

- the remainder of the relative clause instructs us to check that the *u*-farmer owns the *u'*-donkey



3. Dynamic Semantics

4. Every^{*u*} farmer who owns a^{*u'*} donkey beats it_{*u'*}.

	<i>u</i>	<i>u'</i>
Sequence (assignment) <i>i</i> ₁ '		
		

If  owns , we move on to the remainder of sentence (4).

3. Dynamic Semantics

4. Every^{*u*} farmer who owns a^{*u'*} donkey beats it_{*u'*}.

	<i>u</i>	<i>u'</i>
Sequence (assignment) <i>i</i> ₁ '		
		

If  does not own , we discard sequence *i*₁'. Then, we go back to assignment *i*₁ and construct a different sequence *i*₁' that stores a different donkey relative to dref *u'*, e.g.  .

3. Dynamic Semantics

4. Every^{*u*} farmer who owns a^{*u'*} donkey beats it_{*u'*}.

	<i>u</i>	<i>u'</i>
Sequence (assignment) <i>i</i> ₁ '		

Suppose that  actually owns  .

3. Dynamic Semantics

4. Every^{*u*} farmer who owns a^{*u'*} donkey beats it_{*u'*}.

	<i>u</i>	<i>u'</i>
Sequence (assignment) <i>i</i> ₁ '		
		

- the remainder of the donkey sentence in (4) instructs us to check that the *u*-farmer  beats the *u'*-donkey 

3. Dynamic Semantics

4. Every^{*u*} farmer who owns a^{*u'*} donkey beats it_{*u'*}.

	<i>u</i>	<i>u'</i>
Sequence (assignment) <i>i</i> ₁ '		
		

If the *u*-farmer  does not beat the *u'*-donkey , then the quantification in (4) is false. The update fails.

3. Dynamic Semantics

4. Every^{*u*} farmer who owns a^{*u'*} donkey beats it_{*u'*}.

	<i>u</i>	<i>u'</i>
Sequence (assignment) <i>i</i> ₁ '		
		

If the *u*-farmer  beats the *u'*-donkey , then we move on to check whether  beats a different *u'*-donkey  that  owns.

And when we are done with the *u*-farmer  and **all his donkeys**, we move on to examine a different farmer  and so on and so forth.

3. Dynamic Semantics

4. Every^{*u*} farmer who owns *a^{*u'*} donkey* beats *it_{*u'*}*.

Crucially, information states (sequences of individuals) that are incrementally updated in discourse enable us to pass anaphoric information

FROM: the restrictor of the quantification

... *u* is a farmer who owns *a^{*u'*} donkey* ...

TO: the nuclear scope of the quantification

... *u* beats *it_{*u'*}*.

3. Dynamic Semantics

There are several ways to account for the contrast between indefinites, which exhibit binding without c-command (see (10)), and quantifiers, which don't (see (9)).

9. #Every^{*u*} boy who read *every^{*u'*} Harry Potter book* recommended *it_{*u'*}* to his_{*u*} friends.
10. Every^{*u*} boy who read *a^{*u'*} Harry Potter book* recommended *it_{*u'*}* to his_{*u*} friends.

Traditionally (Kamp 1981 and Heim 1982), we discard all the temporarily constructed *u'*-sequences after we are done checking that, for a given boy *u*, the update contributed by the relative clause *u read every^{*u'*} Harry Potter book* goes through.

Therefore, the pronoun *it_{*u'*}* fails to refer because there's no pigeon in the *u'*-pigeonhole. More exactly: there is no *u'*-pigeonhole.

4. Plural Discourse Reference

The whole story, however, is not so simple (it never is...).

Consider again **multiple singular** donkey anaphora.

5. Every^u person who buys a^{u'} book on [amazon.com](https://www.amazon.com) and has a^{u''} credit card uses it_{u''} to pay for it_{u'}.
6. Every^u boy who bought a^{u'} Christmas gift for a^{u''} girl in his class asked her_{u''} deskmate to wrap it_{u'}.

4. Plural Discourse Reference

5. Every^u person who buys a^{u'} book on [amazon.com](https://www.amazon.com) and has a^{u''} credit card uses it_{u''} to pay for it_{u'}.

(5) is interpreted as:

every book buyer and credit-card owner is such that ...

for **every** book that s/he buys on [amazon.com](https://www.amazon.com),

there is **SOME** credit card that s/he has and that s/he uses to pay for the book.

4. Plural Discourse Reference

Thus, there's one complicating factor – the quantificational force of donkey anaphora can vary. It can be:

- **maximal**, i.e. interpreted as an **every**

4. Every^u farmer who owns **a^{u'} donkey** beats **it_{u'}**.

every donkey owning farmer beats **every** donkey s/he owns.

- **non-maximal**, i.e. interpreted as a **some**

12. Every^u driver who had **a^{u'} dime** put **it_{u'}** in the meter.

every dime owning driver put **some** dime s/he had in the meter.

(Pelletier and Schubert 1989)

4. Plural Discourse Reference

5. Every^u person who buys a^{u'} book on [amazon.com](https://www.amazon.com) and has a^{u''} credit card uses it_{u''} to pay for it_{u'}.

And two instances of donkey anaphora can have different quantificational forces within the same sentence.

(5) is a **mixed maximal & non-maximal** donkey sentence since it is interpreted as:

for **every** (maximal) book that any credit-card owner buys on [amazon.com](https://www.amazon.com),

there is **some** (non-maximal) credit card that s/he uses to pay for the book.

4. Plural Discourse Reference

5. Every^u person who buys a^{u'} book on [amazon.com](https://www.amazon.com) and has a^{u''} credit card uses it_{u''} to pay for it_{u'}.
4. Every^u farmer who owns a^{u'} donkey beats it_{u'}.

The previous recipe for evaluating the donkey sentence in (4) is not going to work for (5) because ...

... in (4), we check every donkey owning farmer  ,  ,  etc.
and, for each such farmer, **every** donkey  ,  ,  etc.
that s/he owns,

... while in (5) we check every book, but only **some** credit cards.

4. Plural Discourse Reference

5. Every^{*u*} person who buys a^{*u'*} book on [amazon.com](https://www.amazon.com) and has a^{*u''*} credit card uses it_{*u''*} to pay for it_{*u'*}.

One strategy would be to allow for sets of 'pigeons' (i.e. sets of atomic individuals, i.e. non-atomic individuals) in each pigeonhole.

That is, we allow for plural reference, for example:

	for a person <i>u</i>	store in <i>u'</i> all the books <i>u</i> bought	store in <i>u''</i> some of <i>u</i> 's credit cards
	<i>u</i>	<i>u'</i>	<i>u''</i>
Sequence <i>i</i>			

4. Plural Discourse Reference

5. Every^{*u*} person who buys a^{*u'*} book on [amazon.com](https://www.amazon.com) and has a^{*u''*} credit card uses *it*_{*u''*} to pay for *it*_{*u'*}.

	for a person <i>u</i>	store in <i>u'</i> all the books <i>u</i> bought	store in <i>u''</i> some of <i>u</i> 's credit cards
	<i>u</i>	<i>u'</i>	<i>u''</i>
Sequence <i>i</i>			

Then, check that:

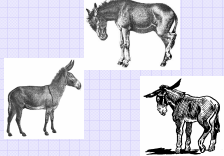
- for any *u'* book, there is a *u''* card such that: *u* paid for *u'* with *u''*
- for any *u''* card, there is a *u'* book such that: *u* paid for *u'* with *u''*

4. Plural Discourse Reference

This strategy, however, incorrectly predicts that (11) is felicitous:

11. #Every^{*u*} farmer who owns *a^{*u'*}* donkey gathers *it_{*u'*}* around the fire at night.

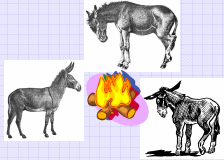
If we store in *u'* all the three donkeys that farmer John has...

	<i>u</i>	<i>u'</i>
Sequence <i>i</i>		

4. Plural Discourse Reference

11. #Every^{*u*} farmer who owns *a^{*u'*}* donkey gathers *it_{*u*}* around the fire at night.

... we should be able to gather them around the fire.

	<i>u</i>	<i>u'</i>
Sequence <i>i</i>		

4. Plural Discourse Reference

Also, we make incorrect predictions...

... with respect to the interpretation of the other multiple singular donkey sentence:

6. Every^u boy who bought a^{u'} Christmas gift for a^{u''} girl in his class asked her_{u''} deskmate to wrap it_{u'}.

4. Plural Discourse Reference

6. Every^{*u*} boy who bought a^{*u'*} Christmas gift for a^{*u''*} girl in his class asked her_{*u''*} deskmate to wrap it_{*u'*}.

In (6), both instances of donkey anaphora are maximal:

we consider every Christmas gift and every girl.

The restrictor of the quantification introduces a **dependency** between the set of *u'*-gifts and the set of *u''*-girls:

each *u'*-gift is correlated with the *u''*-girl it was bought for.

4. Plural Discourse Reference

6. Every^{*u*} boy who bought a^{*u'*} Christmas gift for a^{*u''*} girl in his class asked her_{*u''*} deskmate to wrap it_{*u'*}.

The nuclear scope of the quantification retrieves not only the two sets of objects,
but also the **structure** associated with them (i.e. the dependency between them):

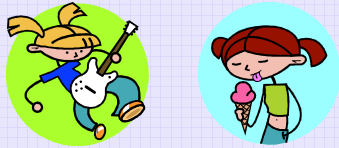
each *u'*-gift was wrapped by the deskmate of the *u''*-girl that the gift was bought for.

We have donkey anaphora to **structure** in addition to donkey anaphora to objects / values.

4. Plural Discourse Reference

6. Every^{*u*} boy who bought a^{*u'*} Christmas gift for a^{*u''*} girl in his class asked her_{*u''*} deskmate to wrap it_{*u'*}.

Now consider what happens if we store sets in pigeonholes:

	for a boy <i>u</i>	store in <i>u'</i> all the gifts <i>u</i> bought for some <i>u''</i> -girl	store in <i>u''</i> all the girls for which <i>u</i> bought some <i>u'</i> -gift
	<i>u</i>	<i>u'</i>	<i>u''</i>
Sequence <i>i</i>			

4. Plural Discourse Reference

	for a boy u	store in u' all the gifts u bought for some u'' -girl	store in u'' all the girls for which u bought some u' -gift
	u	u'	u''
Sequence i		 	 

We do **not** store the **structure** associated with these sets ...

- ... namely the fact that little Johnny  bought
- the first gift  for Megan 
 - the second gift  for Gaby 

4. Plural Discourse Reference

6. Every^u boy who bought a^{u'} Christmas gift for a^{u''} girl in his class asked her_{u''} deskmate to wrap it_{u'}.

But this **structure**,

i.e. the **dependency** between gifts and girls,

needs to be **semantically** encoded and cannot be left underspecified (and for pragmatics to further specify) – in contrast with (3), where it can be underspecified.

3. John bought a^u gift for every^{u'} girl in his class and asked them_{u'} / the_{u'} girls to wrap them_u / the_u gifts.

4. Plural Discourse Reference

6. Every^u boy who bought a^{u'} Christmas gift for a^{u''} girl in his class asked her_{u''} deskmate to wrap it_{u'}.

To see that the gift-girl dependency in (6) needs to be semantically encoded (and is not left for pragmatics to specify),

consider the following scenario...

4. Plural Discourse Reference

6. Every^u boy who bought a^{u'} Christmas gift for a^{u''} girl in his class asked her_{u''} deskmate to wrap it_{u'}.

Suppose (as we just did) that little Johnny buys two gifts:

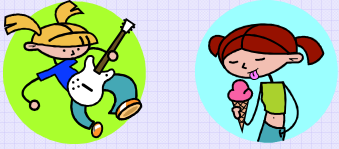
- one  for Megan 
- the other  for Gaby 

Moreover, the two girls are deskmates.

(6) is true if little Johnny asked Megan to wrap Gaby's gift and Gaby to wrap Megan's gift.

(6) is not true if little Johnny asked each girl to wrap her own gift.

4. Plural Discourse Reference

	u	u'	u''
Sequence i			

But if we have only plural reference, i.e. we store the sets of atomic individuals, but not the **structure** associated with these sets ...

we incorrectly predict that (6) is true even in the second situation in which each girl wrapped her own gift.

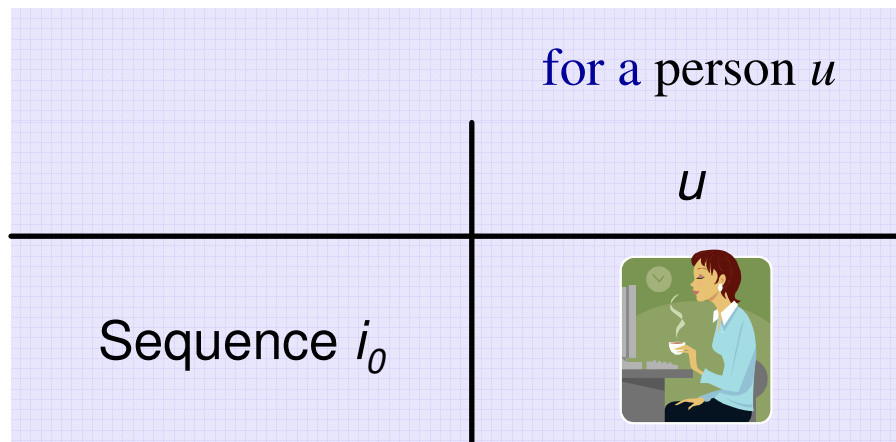
4. Plural Discourse Reference

The solution.

- allow for **plural discourse reference** over and above plural reference
- model plural discourse reference as **plural information states**
- that is, an information state is **a set of sequences** instead of only one sequence – and we incrementally update in discourse an entire such a set of sequences

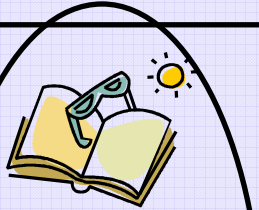
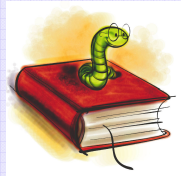
4. Plural Discourse Reference

5. Every^{*u*} person who buys a^{*u'*} book on [amazon.com](https://www.amazon.com) and has a^{*u''*} credit card uses *it*_{*u''*} to pay for *it*_{*u'*}.



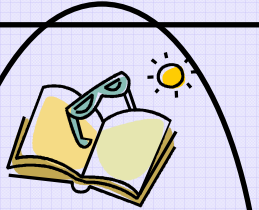
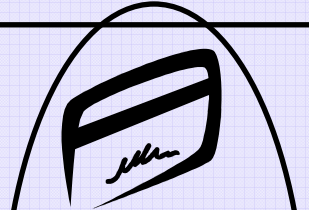
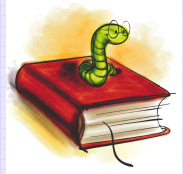
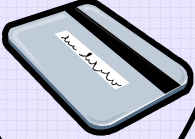
4. Plural Discourse Reference

5. Every^{*u*} person who buys a^{*u'*} book on [amazon.com](https://www.amazon.com) and has a^{*u''*} credit card uses it_{*u''*} to pay for it_{*u'*}.

	for a person <i>u</i>	store in <i>u'</i> all the books <i>u</i> bought
	<i>u</i>	<i>u'</i>
Sequence <i>i</i> ₁		
Sequence <i>i</i> ₂		
Sequence <i>i</i> ₃		

4. Plural Discourse Reference

5. Every^{*u*} person who buys a^{*u'*} book on [amazon.com](https://www.amazon.com) and has a^{*u''*} credit card uses *it*_{*u''*} to pay for *it*_{*u'*}.

	for a person <i>u</i>	store in <i>u'</i> all the books <i>u</i> bought	store in <i>u''</i> some of <i>u</i> 's credit cards
	<i>u</i>	<i>u'</i>	<i>u''</i>
Sequence <i>i</i> ₁ '			
Sequence <i>i</i> ₂ '			
Sequence <i>i</i> ₃ '			

4. Plural Discourse Reference

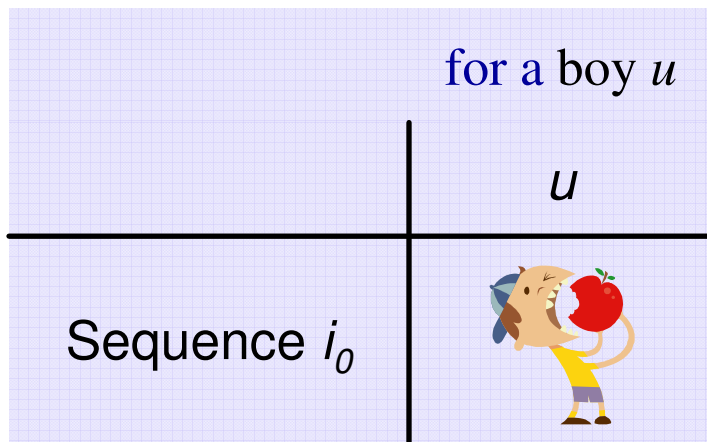
5. Every^{*u*} person who buys a^{*u'*} book on [amazon.com](https://www.amazon.com) and has a^{*u''*} credit card uses *it*_{*u''*} to pay for *it*_{*u'*}.

for each sequence (i.e. each row in the matrix), check that *u* paid for the *u'*-book in that sequence with the *u''*-card in that sequence

	<i>u</i>	<i>u'</i>	<i>u''</i>
Sequence <i>i</i> ₁ '			
Sequence <i>i</i> ₂ '			
Sequence <i>i</i> ₃ '			

4. Plural Discourse Reference

6. Every^{*u*} boy who bought a^{*u'*} Christmas gift for a^{*u''*} girl in his class asked her_{*u''*} deskmate to wrap it_{*u'*}.



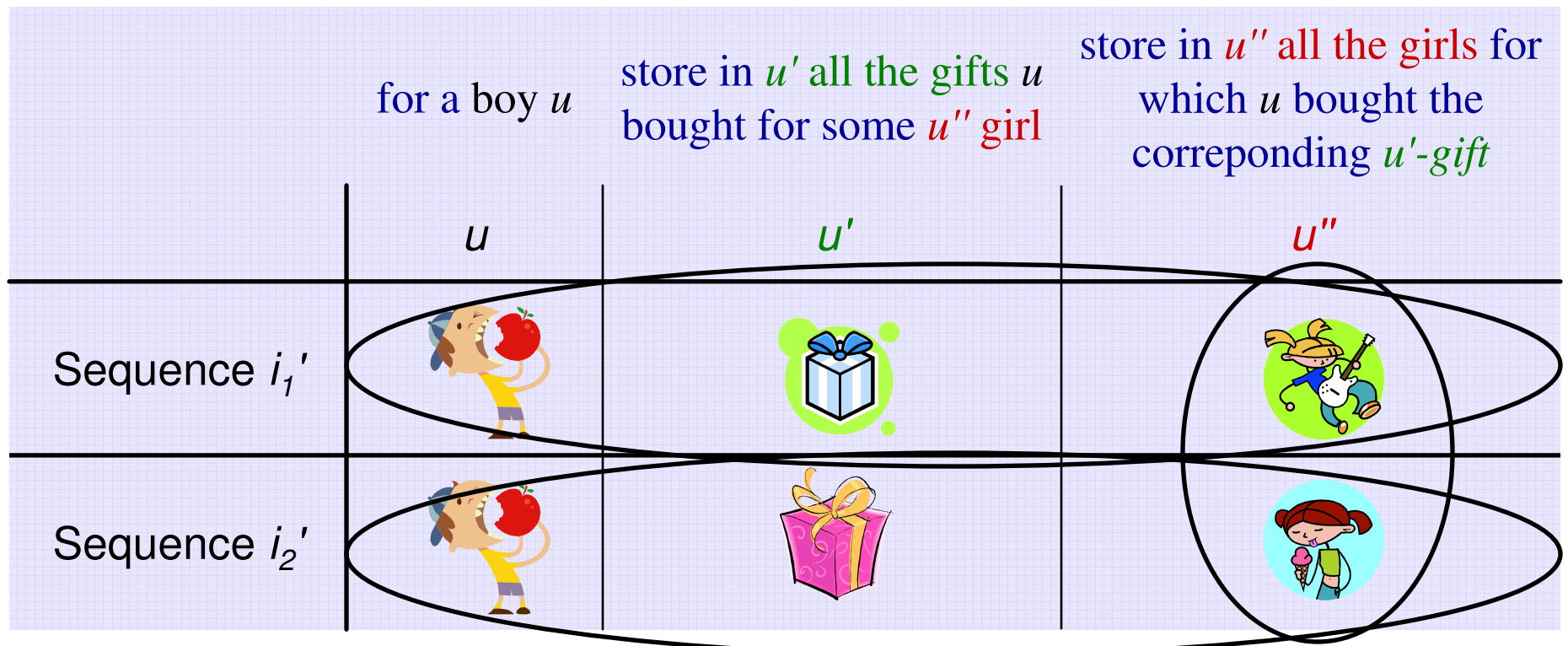
4. Plural Discourse Reference

6. Every^{*u*} boy who bought a^{*u'*} Christmas gift for a^{*u''*} girl in his class asked her_{*u''*} deskmate to wrap it_{*u'*}.

	for a boy <i>u</i>	store in <i>u'</i> all the gifts <i>u</i> bought for some <i>u''</i> girl
	<i>u</i>	<i>u'</i>
Sequence <i>i</i> ₁		
Sequence <i>i</i> ₂		

4. Plural Discourse Reference

6. Every^{*u*} boy who bought a^{*u'*} Christmas gift for a^{*u''*} girl in his class asked her_{*u''*} deskmate to wrap it_{*u'*}.



4. Plural Discourse Reference

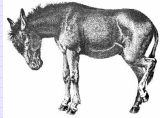
6. Every^{*u*} boy who bought a^{*u'*} Christmas gift for a^{*u''*} girl in his class asked her_{*u''*} deskmate to wrap it_{*u'*}.

for each sequence (i.e. each row the matrix), check that *u* asked the deskmate of the *u''*-girl in that sequence to wrap the *u'*-gift in that sequence

	<i>u</i>	<i>u'</i>	<i>u''</i>
Sequence <i>i</i> ₁ '			
Sequence <i>i</i> ₂ '			

4. Plural Discourse Reference

11. #Every^{*u*} farmer who owns *a^{*u'*} donkey* gathers *it_{*u'*}* around the fire at night.

	for a farmer <i>u</i>	store in <i>u'</i> all the donkeys <i>u</i> has – one <i>u'</i> -donkey at a time!
	<i>u</i>	<i>u'</i>
Sequence <i>i</i> ₁		
Sequence <i>i</i> ₂		
Sequence <i>i</i> ₃		

4. Plural Discourse Reference

11. #Every^{*u*} farmer who owns a^{*u'*} donkey gathers it_{*u'*} around the fire at night.

for each sequence, check that *u* gathers the *u'*-donkey in that sequence around the fire at night – hence: infelicity!

	<i>u</i>	<i>u'</i>
Sequence <i>i</i> ₁		
Sequence <i>i</i> ₂		
Sequence <i>i</i> ₃		

4. Plural Discourse Reference

Finally,

allowing for **plural reference** over and above plural discourse reference ...

enables us to account for the intuitive parallel between singular and plural donkey anaphora ...

for example ...

4. Plural Discourse Reference

13. Every^u parent who gives a^{u'} balloon / three^{u'} balloons to two^{u''} boys expects them_{u''} to end up fighting (each other) for it_{u'} / them_{u'}.

(based on an example due to Maria Bittner, p.c.)

... which is parallel to ...

6. Every^u boy who bought a^{u'} Christmas gift for a^{u''} girl in his class asked her_{u''} deskmate to wrap it_{u'}.

4. Plural Discourse Reference

6. Every^u boy who bought a^{u'} Christmas gift for a^{u''} girl in his class asked her_{u''} deskmate to wrap it_{u'}.

	u	u'	u''
Sequence i_1'			
Sequence i_2'			

13. Every^u parent who gives a^{u'} balloon / three^{u'} balloons to two^{u''} boys expects them_{u''} to end up fighting (each other) for it_{u'} / them_{u'}.

	u	u'	u''
Sequence i_1'			
Sequence i_2'			

The matrix on the first slide should be intelligible now.

Plural Info State I	...	U	U'	...
i_1	...	x_1 (=jane $\oplus$ john)	y_1 (=mary)	...
i_2	...	x_2 (=bill $\oplus$ gaby)	y_2 (=sue $\oplus$ joe)	...
i_3	...	x_3 (=megan)	y_3 (=matt $\oplus$ rob)	...
...	...	...	...	...

Individuals, possibly non-atomic (plural reference): $\{x_1, x_2, x_3, \dots\}, \{y_1, y_2, y_3, \dots\}$ etc.	Quantificational dependencies between individuals (plural discourse reference): $\{ \langle x_1, y_1 \rangle, \langle x_2, y_2 \rangle, \langle x_3, y_3 \rangle, \dots \}$ etc.
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5. Instead of Conclusion

The notion of plural discourse reference and the resulting compositional dynamic system (dubbed Plural Compositional DRT – PCDRT) ...

... can be straightforwardly generalized to capture temporal anaphora, event anaphora, modal anaphora etc.

For example, we can capture the interaction between anaphora to individuals, modal anaphora and the entailment particle *therefore* in the following discourse based on Thomas Aquinas.

5. Instead of Conclusion

The original Aquinas discourse is:

[A] man cannot live without joy. Therefore, when he is deprived of true spiritual joys, it is necessary that he become addicted to carnal pleasures.

But we will look at the more explicit (and more awkward) variant:

14. **a.** If a man is alive, he must find something pleasurable.
- b.** Therefore, if he does not have a spiritual pleasure, he must have a carnal pleasure.

5. Instead of Conclusion

14. **a.** If **a man** is alive, **he** must find something pleasurable.
b. Therefore, if **he** does not have a spiritual pleasure, **he** must have a carnal pleasure.

The questions arises: who is *he* in (14b)?

Also: how can we compositionally account for the interpretation of the argument in (14) ...

... given that the conditional in (14b), i.e. the conclusion, is in fact interpreted as duplicating the antecedent of the conditional in (14a), i.e. the premise? ...

5. Instead of Conclusion

14. **a.** If **a man** is alive, **he** must find something pleasurable.
b. Therefore, if **he** does not have a spiritual pleasure, **he** must have a carnal pleasure.

That is, the argument in (14) is interpreted as ...

- 14'. **a.** If **a man** is alive, **he** must find something pleasurable.
b. Therefore, if **a man is alive and he** does not have a spiritual pleasure, **he** must have a carnal pleasure.

5. Instead of Conclusion

14. a. If a man is alive, he must find something pleasurable.
b. Therefore, if he does not have a spiritual pleasure, he must have a carnal pleasure.

Given the notion of plural discourse reference, i.e. of **structured** anaphora,

we can analyze the Aquinas discourse as a network of structured anaphoric connections.

The meaning (and validity) of the argument emerges as a consequence of the intertwined individual-level and modal anaphora.

5. Instead of Conclusion

Moreover, we can explicitly and systematically capture the anaphoric and quantificational parallels between the modal and individual domains...

... e.g. the parallel between (15) (Karttunen 1976) and (16) (based on an example in Roberts 1989) below.

(15) a. Harvey courts a girl at every convention.
b. She always comes to the banquet with him.

(16) a. A wolf might come in.
b. It would eat Harvey first.

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